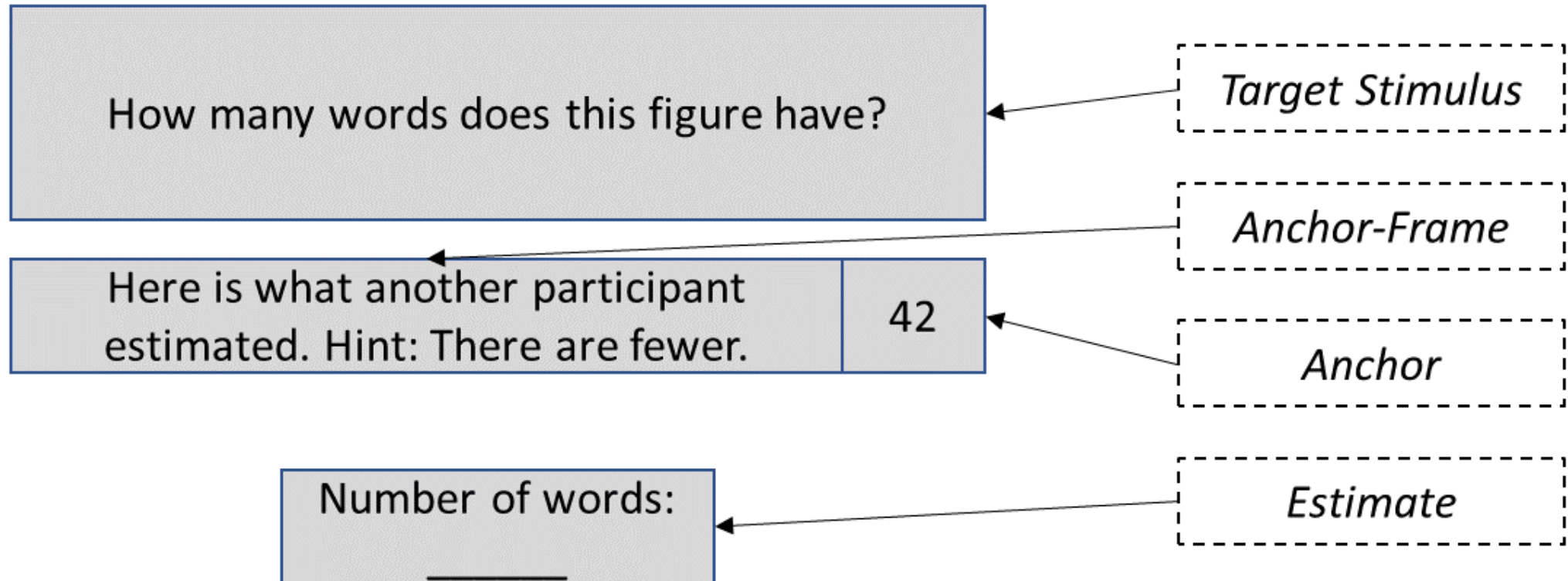


The Open Anchoring Quest

Explaining variance of the
heterogeneous but large anchoring effects



Anchoring Effects



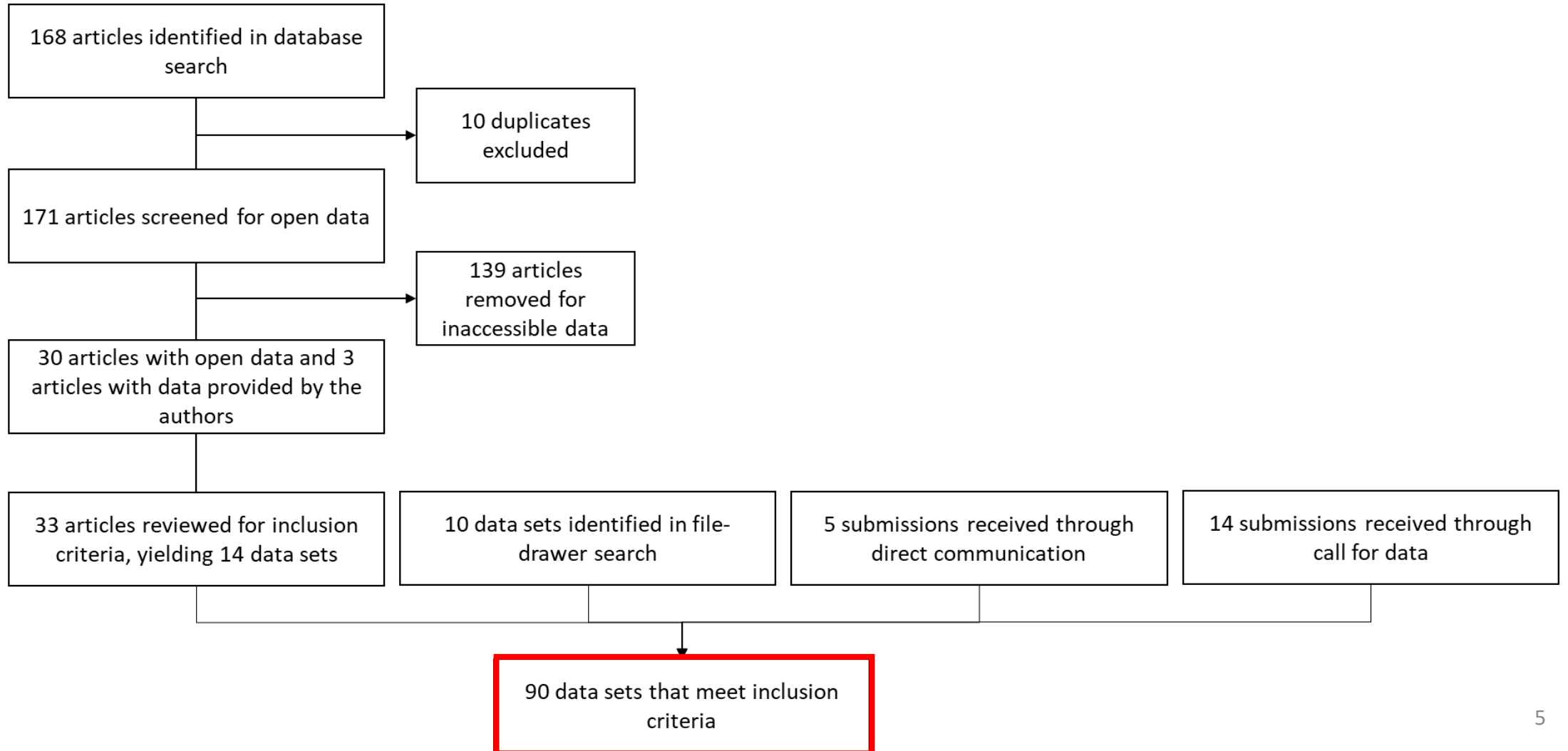
State of the field

- Robust anchoring effects
- Seminal theories have been facing replication failures
- Almost no robust moderator findings
- Heterogeneous paradigms
- For researchers to come up with an accurate theory/model, contradictions have to be resolved
- We conducted a meta-analysis to resolve the contradictions and provide the basis for theory development

Domain	Moderator	Insufficient adjustment	Selective accessibility	Scale Distortion Theory	Past Findings
Motivation	Incentives for accurate estimates	Incentives decrease deliberative adjustment	—	—	- Incentives decrease deliberate adjustment: Epley & Gilovich, 2005; LeBoeuf & Shafir, 2009; Meub et al., 2013 - No difference: Enke et al., 2021; L. Li et al., 2021; Simmons et al., 2010; Tversky & Kahneman, 1974; Wilson et al., 1996
Participant traits	Expertise	—	—	—	- Smaller effects for experts: Bystranowski et al., 2021; Englich & Soder, 2009; Orr & Guthrie, 2006 - Larger effects for experts: Townson, 2019
	Age	—	—	—	—
	Gender	—	—	—	—
Memory	Priming	—	Anchors prime concepts that are consistent with the anchor	—	- SA: Mussweiler & Strack, 1999b - Null: Bahnik, 2021a; Harris et al., 2019; Street & Kingstone, 2017; Wong & Kwong, 2000
Process characteristics	Adjustment steps	People adjust in steps	—	—	- IA: Epley & Gilovich, 2001; Frech et al., 2020
Paradigm features	Adjustment direction	—	—	—	—
	Lab vs. online experiments	—	—	—	- Larger variance in nonlab experiment: L. Li et al., 2021, Table 3
	Time pressure	Time pressure leads to stronger anchoring (Epley & Gilovich, 2006, p. 312-313)	More thorough processing leads to more anchoring (Englich & Soder, 2009)	—	- IA: Lieder et al., 2018b; Yik et al., 2019, Study 3 - Null: Chaxel, 2014
	Anchor extremeness	People do not adjust away from anchors that fall within their range of plausible values	Extreme primes can lead to contrast effects	—	- Nonlinearity: Chapman & Johnson, 1994; Thorsteinson, 2011 - Possible inverted u-shape: Bystranowski et al., 2021, p. 13; Townson, 2019
	Description of stimulus	—	—	—	- Audio/video > briefly written > written in detail: Bystranowski et al., 2021
	Precision of anchor (e.g., 49.98 vs. 50.00)	Precise anchors have larger <i>afforts</i>	—	Precise anchors have larger <i>afforts</i>	- IA: Frech et al., 2019, 2020; Janiszewski & Uy, 2008; Loschelder et al., 2014; Mason et al., 2013
	Comparative question	—	Consideration of the anchor in the comparative question increases anchoring effects	—	—
	Scale type	—	—	—	- No difference: Bystranowski et al., 2021, p. 13
	Anchor manipulation (continuous vs. high-low)	—	—	—	- No difference: Bystranowski et al., 2021, p. 14
	Type of anchoring task (estimating weight, height, price, etc.)	—	—	—	—
	Relevance of anchor	Random anchors have smaller effects than nonrandom anchors	Applicability of information leads to stronger anchoring effects (Strack & Mussweiler,	Random anchors have smaller effects than nonrandom anchors	- Smaller effects for irrelevant anchors: L. Li et al., 2021, Table 3; Frederick et al., 2014; Schweickart et al., 2020; Sugden et al., 2013; Townson, 2019; No difference: Englich et al., 2006, p. 194; Bystranowski et al., 2021, p. 14
Study characteristics	Published in peer-reviewed journal vs. the rest	—	—	—	- Publication bias: Bystranowski et al., 2021
	Preregistered	—	—	—	—
	Year of publication	—	—	—	- No publication bias: L. Li et al., 2021

The OpAQ-dataset:

Aggregation of anchoring datasets (trial-level)



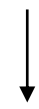


OpAQ: Unus pro omnibus

References	Studies	N	Items	Trials
51	90	19585	398	82279

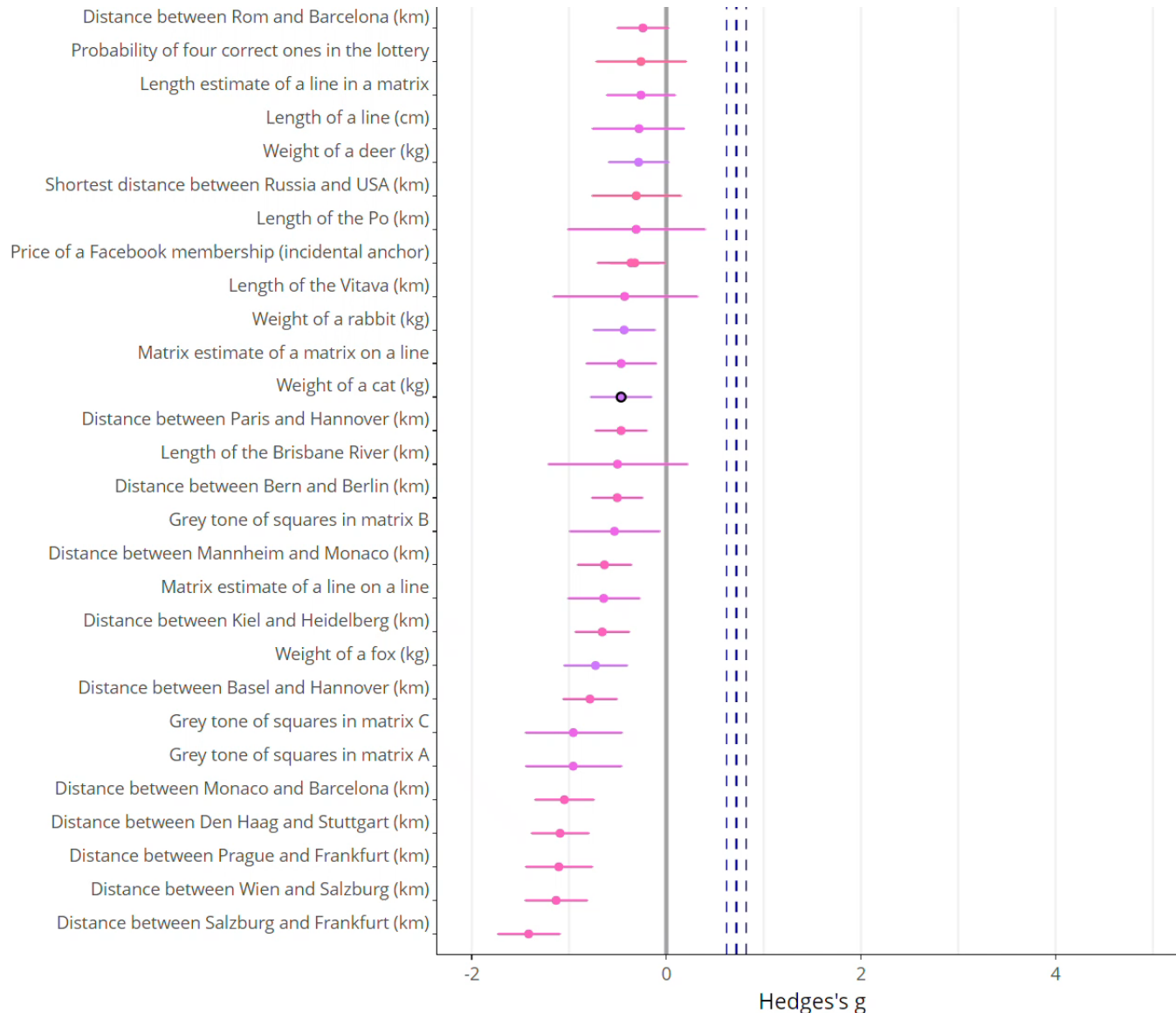
We have been accumulating trial-level data for all open anchoring datasets and reaching out to authors for their original data

This is what this looks like



participant_id	sex	age	anchoring_item	true_value	anchor	anchorhigh	estimate	tasktype	direction	incentive	comparative_question	experiment_type
1001			Length of the Rio Japura (km)	2816	1408	0	5000	length	1	0	0	1
1001			Length of the Lena (km)	4294	2147	0	3500	length	1	0	0	1
1001			Length of the Vistula (km)	1047	524	0	1000	length	1	0	0	1
1001			Length of the Po (km)	652	326	0	560	length	1	0	0	1
1001			Length of the Ems (km)	371	186	0	200	length	1	0	0	6

Anchoring Effect Sizes



Current effect size

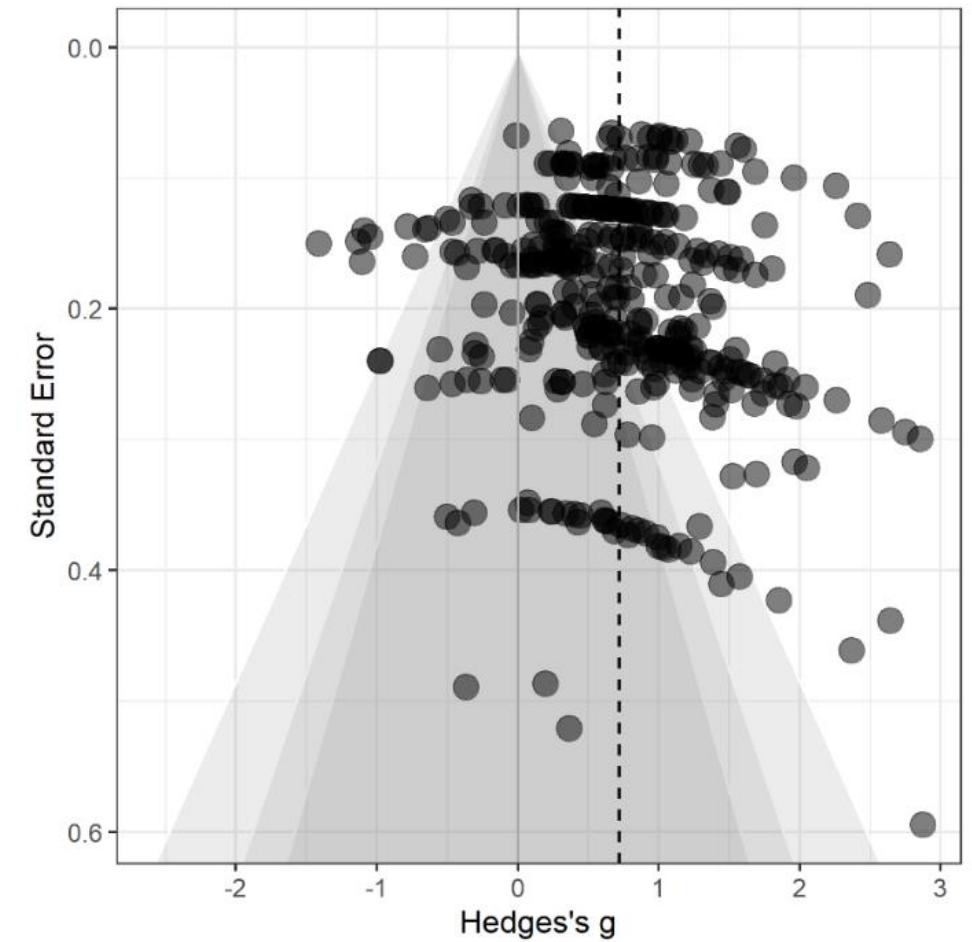
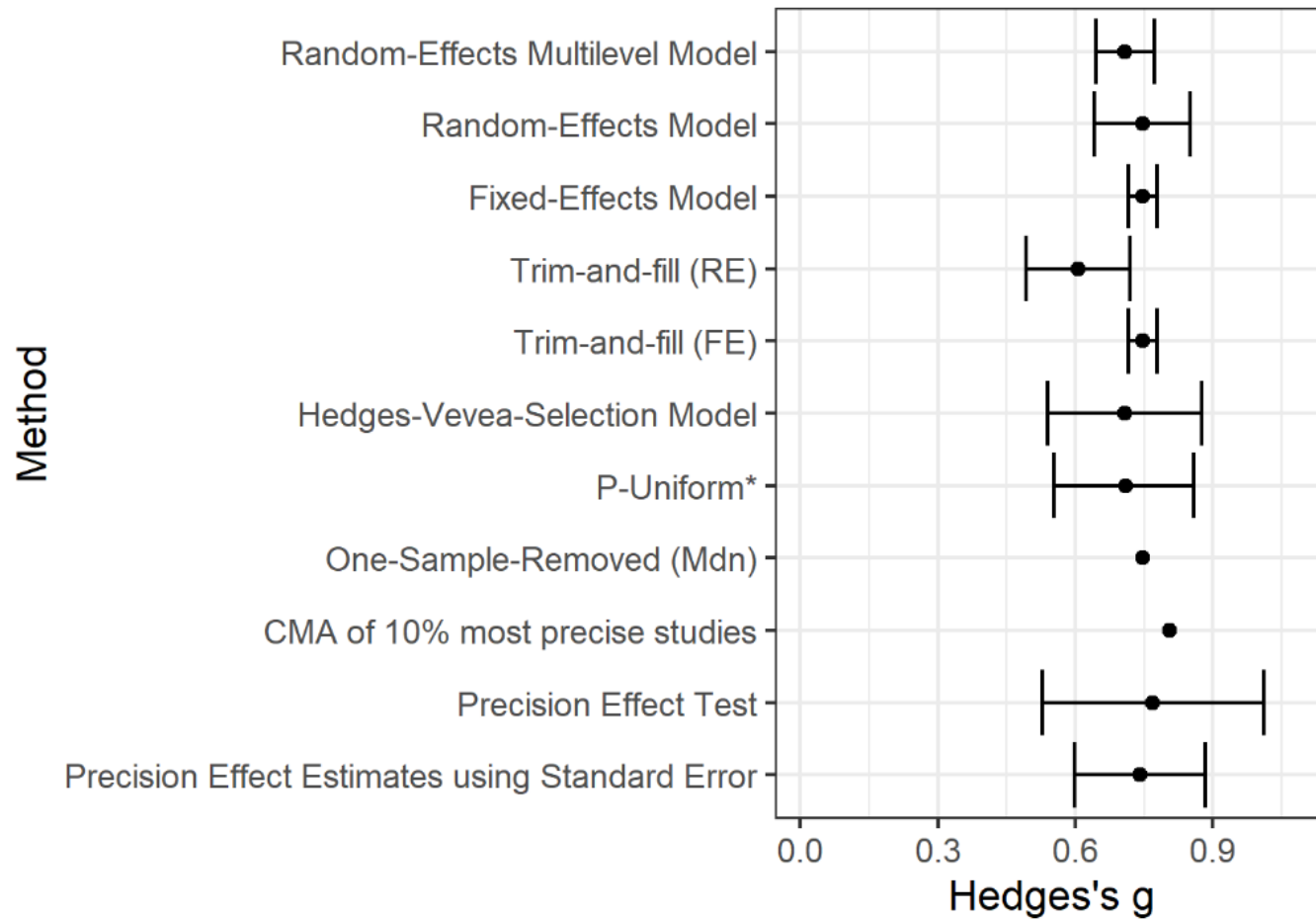
$g = 0.720, p < .001$

95% CI = [0.619, 0.821]

95% PI = [-0.192, 1.632]

$\sigma^2 = 0.213, N_{total} = 17488, k = 405$

Data Quality



Data Quality

Moderators

Pre-Registered

Show 10 entries

Search:

Moderator_Levels	g	ci_lower	ci_upper	k
no	0.702	0.577	0.826	206
yes	0.754	0.585	0.924	199

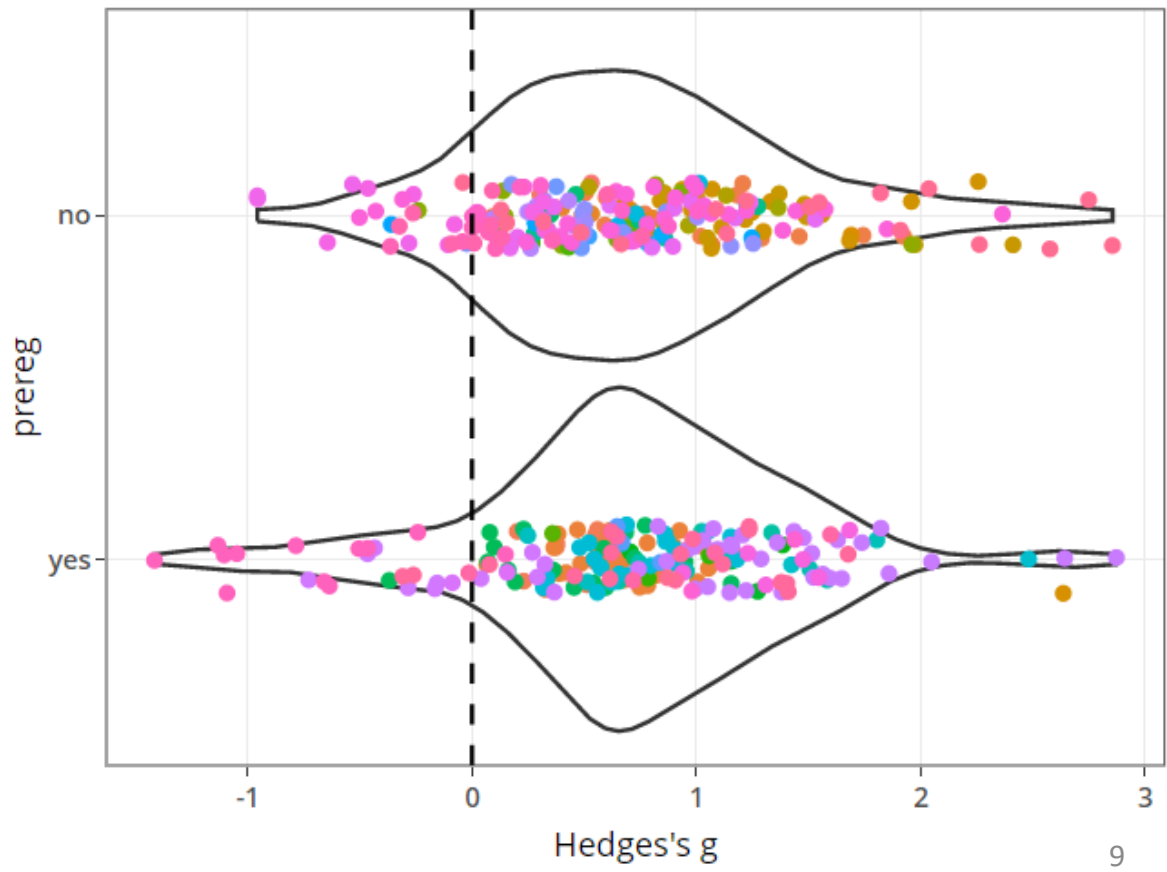
Showing 1 to 2 of 2 entries

Previous

1

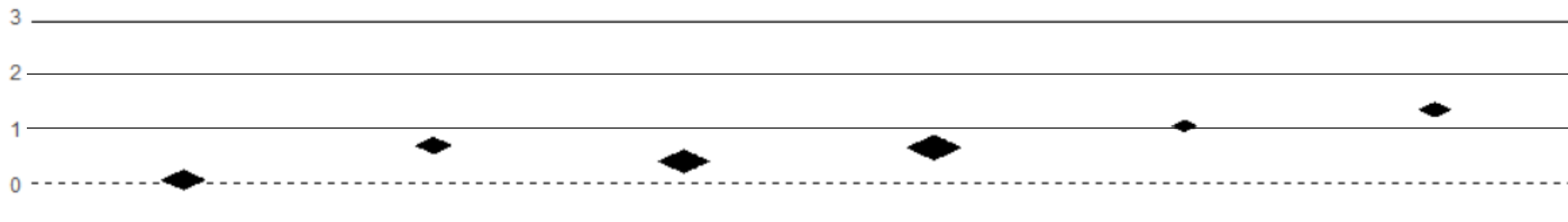
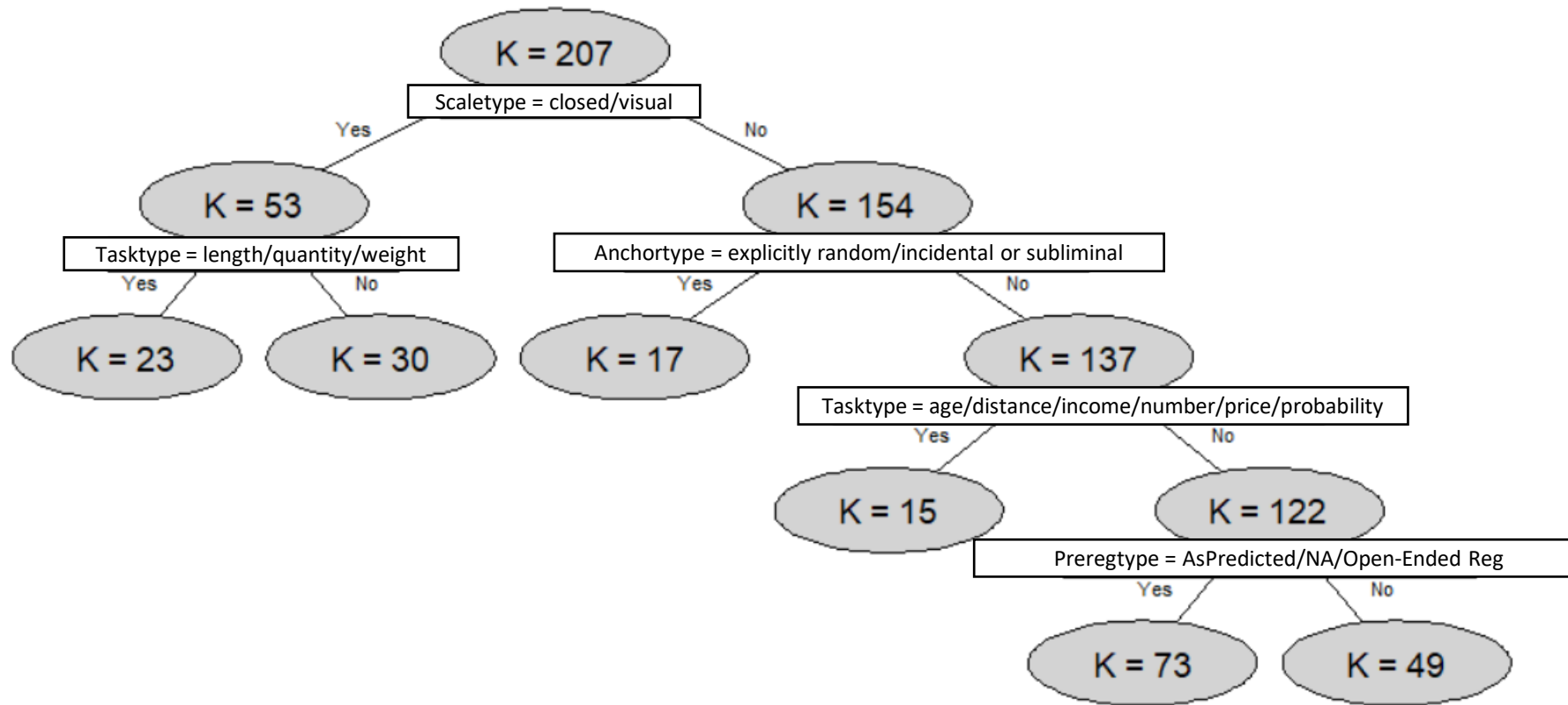
Next

The effect of **prereg** on anchoring effect sizes is **not** significant at the 5% level. Test of moderators: $F(1, 403) = 0.24, p = 0.622$.



Moderator	Past findings	This MA
Incentives for accurate estimates	- <i>Incentives decrease deliberate adjustment: Epley & Gilovich, 2005; LeBoeuf & Shafir, 2009; Meub et al., 2013</i> - No difference: Enke et al., 2021; L. Li et al., 2021; Simmons et al., 2010; Tversky & Kahneman, 1974; Wilson et al., 1996	No difference
Age	—	No relationship
Gender	—	No relationship
Adjustment direction	—	No difference
Lab vs. online experiments	- <i>Larger variance in nonlab experiment: L. Li et al., 2021, Table 3</i>	No difference
Anchor extremeness	- Nonlinearity: Chapman & Johnson, 1994; Thorsteinson, 2011 - <i>Possible inverted u-shape: Bystranowski et al., 2021, p. 13; Townson, 2019</i>	Positive and nonlinear relationship between extremeness and effect size
Description of stimulus	- <i>Audio/video > brief written description > detailed written description; Bystranowski et al., 2021</i>	Image/audio/video < brief written description < other
Comparative question	—	Larger effects when a comparative question was present (probably due to incidental anchors not having CQ)
Scale type	- <i>No difference: Bystranowski et al., 2021, p. 13</i>	Open > visual > closed
Anchor manipulation (continuous vs. high-low)	- No difference: Bystranowski et al., 2021, p. 14	No difference
Type of anchoring task (estimating weight, height, price, etc.)	—	predictions > quantity > length > temperature > height > date > price [considering only k > 10 studies]
Relevance of the anchor	- Smaller effects for irrelevant anchors: L. Li et al., 2021, Table 3; Frederick et al., 2014; Schweickart et al., 2020; Sugden et al., 2013; Townson, 2019 - <i>No difference: English et al., 2006, p. 194; Bystranowski et al., 2021, p. 14</i>	Relevant > fixed > random
Size of the numbers (e.g., > 100 vs. ~10000)	- Larger effect for larger numbers: Lee & Morewedge, 2021	
Published in peer-reviewed journal vs. the rest	- <i>Publication bias: Bystranowski et al., 2021</i> - No publication bias: L. Li et al., 2021	No difference
Preregistered	—	No difference
Year of publication	- <i>Decreasing effect: L. Li et al., 2021, Figure 3</i>	No relationship

Exploratory Moderator Search



What should an anchoring model look like?

- Formalized & independent of researchers
- Include moderators
 - Anchor-distance
 - Anchor precision
 - Number of anchors
- Model informativeness
 - Of anchor
 - Of target stimulus

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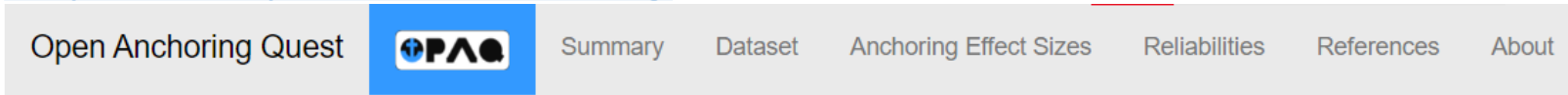
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The OpAQ-App

t1p.de/openanchoring



Welcome to the Open Anchoring Quest!

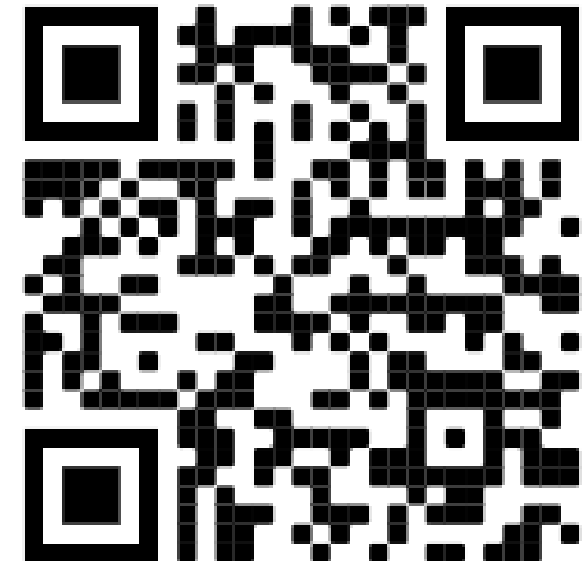
What is OpAQ?

Research on anchoring effects is currently challenged by the lack of reliability of anchoring scores. That is, susceptibility to anchoring for one item is almost completely uncorrelated with susceptibility to anchoring on any other item. With OpAQ, we try to tackle this problem by meta-analytically investigating different measures of anchoring susceptibility as well as factors influencing their reliability. For this purpose, we have created an open, free, and ever-growing dataset already containing thousands of trials from several studies on anchoring effects – and with your support soon even more.

What are your benefits of joining us?

You are very welcome to contribute data from your anchoring experiment(s)! In return, (apart from being rewarded by the good feeling of helping research on anchoring effects to improve) we will list you as a co-author of the OpAQ. If you provide us your data, we can convert it and add it to the OpAQ dataset. You can also use our data curation template and convert your data yourself. Accordingly, we will mention your contribution either as “Resources” or additionally as “Data curation”.

Unpublished datasets as well as data from classroom experiments are also highly appreciated.



Core Contributors

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Resources Contributors

Antunovic, M., Bahník, Š., Barrera, F., Baumeister, R. F., Bermeitinger, C., Bickenbach, S. L. C., Blank, P. A., Blower, F. B. N., Boo, F. L., Boruchowicz, C., Bühler, R. L., Burgmer, P., Cheek, N., N., Dohle, S., Dorsch, L., Dück, M. S., Eliran, H., Fels, S.-A., Frech, M.-L., Freira, L., Friedinger, K., Genschow, O., Häusser, J. A., Henkel, M., Horvath, D., Hügel, J. C., Igna, E. L. E., Intelmann, P., Karg, A. H., Klamar, A., Klusmann, B., Knappe, E., Kraft, P., Kroworsch, M. K., Kühling, S., Lammers, J., Loschelder, D. D., Milstein, N., Molden, D. C., Navajas, J., Norem, J., K., Novak, J. Page, E., Panse, F., Papenmeier, F., Pavlovic, Z., Rebholz, T. R., Rinn, R., Rodgers, S., Röseler, J. J., Sartorio, M., Scheelje, L., Schindler, S., Schreiner, N. B., Schreiter, M. L., Seida, C., Shanks, D. R., Siems, M.-C., Stitz, M., Starkulla, M., Stäglich, M., Thies, K., Thum, E., Undorf, M., Unger, B. D., Vadillo, M. A., Wackershauser-Sablotny, V., Wagner, F. S., Wessel, I., Wolf, H., Zhou, A., Zimdahl, M.

